

Step 03 - Split Before Training

A First Practical Machine Learning Workflow

What this document is for

Step 02 gave the notebook a small visible dataset and separated it into X and y. Step 03 makes the first serious machine-learning habit explicit: split the data before training, so later scores mean something.

By the end of this step, you should have:

- A clear reason for splitting before fitting a model.
- A practical train/test split created from the X and y objects built in Step 02.

1. Where this step sits

We are still not training a model. That happens next. This step prepares the notebook so that, when a model does arrive, it cannot cheat by being judged only on examples it already saw.

Step	What changes
01 - Get the notebook ready	The environment and package checks exist.
02 - Create a tiny dataset	Rows, features, target, X, and y exist.
03 - Split before training	The rows are divided into training and test portions before fit() appears.
04 - Fit a regression model	A model learns from the training portion only.

The rule for this whole step

Split first. Fit later. If the model has already learned from every row, a later test on those same rows is not an independent test.

2. Why split at all?

Training performance answers one question: how well does the model fit examples it was allowed to learn from? Test performance answers a more useful question: how well does the trained model behave on examples held back from training?

Data portion	Used for	What it tells you
Training data	Learning the model from examples.	Whether the model can fit the material it has seen.
Test data	Checking the trained model on held-back examples.	Whether the result looks likely to generalise.

Plain-English version

Do not revise from the answer sheet and then use the same answer sheet as the exam. Training data is revision material. Test data is the held-back exam paper.

3. What about training, validation, and test?

Yes: larger ML workflows often use three sets, not two. The extra set is the validation set. It exists because model development usually involves choices: which model, which settings, which features, which amount of regularisation, and so on.

Set	Main purpose	Typical use
Training	Learn the model parameters.	The data passed into fit().
Validation	Compare choices during development.	Tune hyperparameters, compare candidate models, pick features.
Test	Final-ish held-back check.	Estimate performance after you have stopped making modelling decisions.

Why this document still uses two sets

This toy dataset has only 12 rows. A three-way split would leave each set too small to be meaningful. For this first workflow, use a simple train/test split. Later, when the data is larger or when cross-validation appears, validation becomes more useful.

Important discipline

The validation set is for making choices. The test set is not. If you keep changing the model until the test score looks good, the test set has quietly become part of training-by-feedback.

4. Import the splitting helper only when it is needed

The notebook already has X and y from Step 02. Now it needs one new tool: scikit-learn's `train_test_split` helper. This helper keeps matching X rows and y values together while it splits them.

Why this cell exists

This is the first point in the workflow where splitting is needed. The import does not split anything by itself; it only makes the helper function available.

Code cell

```
from sklearn.model_selection import train_test_split
```

What to look for

The cell should run silently. No model exists yet, and no rows have moved yet.

5. Re-check the objects you are about to split

Before splitting, ask a boring but useful question: do X and y still have the row counts expected from Step 02? This avoids chasing mysterious shape errors later.

Why this cell exists

X should be the evidence table: one row per house and one column per feature. y should be the known answer: one price per house.

Code cell

```
print("X shape:", X.shape)
print("y shape:", y.shape)
```

What to look for

X should have 12 rows and 4 columns. y should have 12 values. The row count must match because each feature row needs one answer.

6. Make the split

Now divide the rows into two groups. The training group will be used to fit the model in Step 04. The test group will be held back until we want an independent-ish check.

Why this cell exists

`test_size=0.33` holds back roughly one third of the rows for testing. `random_state=42` makes the random split repeatable, which is useful while learning and while comparing notes.

Code cell

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.33,
    random_state=42,
)
```

What to look for

The cell should run silently. It creates four new objects. The original X and y still exist; they have not been overwritten.

7. Inspect the split

The names matter. X_train and y_train belong together. X_test and y_test belong together. The feature objects are still two-dimensional; the target objects are still one-dimensional.

Code cell

```
print("X_train:", X_train.shape)
print("X_test: ", X_test.shape)
print("y_train:", y_train.shape)
print("y_test: ", y_test.shape)
```

What to look for

With 12 rows and test_size=0.33, expect 8 training rows and 4 test rows. X_train should be (8, 4), X_test should be (4, 4), y_train should be (8,), and y_test should be (4,).

Checkpoint

In your own words: X_train and y_train are the material the model may learn from. X_test and y_test are held back so we can later ask whether the trained model generalises.

Stop here before Step 04

You now have the four objects a first supervised-learning example needs: X_train, X_test, y_train, and y_test. The next document can introduce a regression model without also explaining splitting at the same time.

Terms to Say Out Loud

Step 03 - Split Before Training

Retrieval target

You do not need a dictionary definition. You should be able to say each term in plain English and connect it to the notebook work.

Term	Say it out loud as...
training data	The rows the model is allowed to learn from.
test data	Held-back rows used to check the trained model later.
train/test split	The act of dividing X and y into training and test portions before fitting.
held-back	Kept away from training so it can be used for a more honest check.
validation set	A development set used to compare choices before the final-ish test check.
test_size	The fraction or count of rows reserved for testing.
random_state	A fixed random seed that makes a split repeatable.
generalise	Behave usefully on examples the model did not train on.